
Binomial Distribution A Bernoulli trial can result in a success with probability p and a failure with probability $q = 1 - p$. Then the probability distribution of the binomial random variable X , the number of successes in n independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

Note that when $n = 3$ and $p = 1/4$, the probability distribution of X , the number of defectives, may be written as

$$b\left(x; 3, \frac{1}{4}\right) = \binom{3}{x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x}, \quad x = 0, 1, 2, 3,$$

Example 5.1: The probability that a certain kind of component will survive a shock test is $3/4$.
Find the probability that exactly 2 of the next 4 components tested survive.

Solution: Assuming that the tests are independent and $p = 3/4$ for each of the 4 tests, we obtain

$$b\left(2; 4, \frac{3}{4}\right) = \binom{4}{2} \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 = \left(\frac{4!}{2! 2!}\right) \left(\frac{3^2}{4^4}\right) = \frac{27}{128}.$$

A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

- (a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?
- (b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment?

Solution: (a) Denote by X the number of defective devices among the 20. Then X follows a $b(x; 20, 0.03)$ distribution. Hence,

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) = 1 - b(0; 20, 0.03) \\ &= 1 - (0.03)^0(1 - 0.03)^{20-0} = 0.4562. \end{aligned}$$

(b) In this case, each shipment can either contain at least one defective item or not. Hence, testing of each shipment can be viewed as a Bernoulli trial with $p = 0.4562$ from part (a). Assuming independence from shipment to shipment

and denoting by Y the number of shipments containing at least one defective item, Y follows another binomial distribution $b(y; 10, 0.4562)$. Therefore,

$$P(Y = 3) = \binom{10}{3} 0.4562^3 (1 - 0.4562)^7 = 0.1602.$$



The mean and variance of the binomial distribution $b(x; n, p)$ are
 $\mu = np$ and $\sigma^2 = npq$.

Multinomial Distribution If a given trial can result in the k outcomes $E_1, E_2, \dots, E_k$ with probabilities $p_1, p_2, \dots, p_k$, then the probability distribution of the random variables $X_1, X_2, \dots, X_k$, representing the number of occurrences for $E_1, E_2, \dots, E_k$ in n independent trials, is

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

with

$$\sum_{i=1}^k x_i = n \text{ and } \sum_{i=1}^k p_i = 1.$$

Example 5.7: The complexity of arrivals and departures of planes at an airport is such that computer simulation is often used to model the “ideal” conditions. For a certain airport with three runways, it is known that in the ideal setting the following are the probabilities that the individual runways are accessed by a randomly arriving commercial jet:

Runway 1: $p_1 = 2/9$,

Runway 2: $p_2 = 1/6$,

Runway 3: $p_3 = 11/18$.

What is the probability that 6 randomly arriving airplanes are distributed in the following fashion?

Runway 1: 2 airplanes,

Runway 2: 1 airplane,

Runway 3: 3 airplanes

Solution: Using the multinomial distribution, we have

$$\begin{aligned} f\left(2, 1, 3; \frac{2}{9}, \frac{1}{6}, \frac{11}{18}, 6\right) &= \binom{6}{2, 1, 3} \left(\frac{2}{9}\right)^2 \left(\frac{1}{6}\right)^1 \left(\frac{11}{18}\right)^3 \\ &= \frac{6!}{2! 1! 3!} \cdot \frac{2^2}{9^2} \cdot \frac{1}{6} \cdot \frac{11^3}{18^3} = 0.1127. \end{aligned}$$

The **Poisson distribution** is used when modeling the number of occurrences of an event within a fixed interval of time, space, or another domain, given the following conditions:

When to Apply the Poisson Distribution:

1.Counting Discrete Events: The distribution applies to the count of discrete events that occur in a fixed interval.

2.Independence of Events: Events occur independently of each other.

3.Constant Average Rate (λ): The average rate of events (λ) in the interval is constant over time or space.

4.Rare Events: The events are relatively rare compared to the total possible occurrences.

5.Single Event per Instant: Two events cannot occur simultaneously or in an infinitesimally small interval.

On average, 3 traffic accidents per month occur at a certain intersection. What is the probability that in any given month at this intersection (a) exactly 5 accidents will occur? (b) fewer than 3 accidents will occur? (c) at least 2 accidents will occur?

(a) Probability of exactly 5 accidents ($P(X = 5)$):

Substitute $\lambda = 3$ and $k = 5$ into the formula:

$$P(X = 5) = \frac{3^5 e^{-3}}{5!}$$

(b) Probability of fewer than 3 accidents ($P(X < 3)$):

This is the sum of probabilities for $X = 0, 1, 2$:

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$$

For $P(X = 0)$:

$$P(X = 0) = \frac{3^0 e^{-3}}{0!} = \frac{1 \cdot 0.0498}{1} = 0.0498$$

For $P(X = 1)$:

$$P(X = 1) = \frac{3^1 e^{-3}}{1!} = \frac{3 \cdot 0.0498}{1} = 0.1494$$

For $P(X = 2)$:

$$P(X = 2) = \frac{3^2 e^{-3}}{2!} = \frac{9 \cdot 0.0498}{2} = 0.2240$$

Total:

$$P(X < 3) = 0.0498 + 0.1494 + 0.2240 = 0.4232$$

Thus, $P(X < 3) \approx 0.4232$ (or 42.32%).

(c) Probability of at least 2 accidents ($P(X \geq 2)$):

This is the complement of $P(X < 2)$, which includes $P(X = 0) + P(X = 1)$:

$$P(X \geq 2) = 1 - P(X < 2)$$

Calculate $P(X < 2)$:

$$P(X < 2) = P(X = 0) + P(X = 1) = 0.0498 + 0.1494 = 0.1992$$

Complement:

$$P(X \geq 2) = 1 - 0.1992 = 0.8008$$

Thus, $P(X \geq 2) \approx 0.8008$ (or 80.08%).

Example 5.17: During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?

Solution: Using the Poisson distribution with $x = 6$ and $\lambda t = 4$ and referring to Table A.2, we have

$$p(6; 4) = \frac{e^{-4}4^6}{6!} = \sum_{x=0}^6 p(x; 4) - \sum_{x=0}^5 p(x; 4) = 0.8893 - 0.7851 = 0.1042.$$

The negative binomial distribution is a generalization of the geometric distribution. Both distributions are related to the binomial distribution, but they differ in how the random variable is defined. The geometric distribution is used when trials are performed until the first success, while the negative binomial distribution is used when trials are performed until a specific number of successes are achieved.

Negative Binomial Distribution If repeated independent trials can result in a success with probability p and a failure with probability $q = 1 - p$, then the probability distribution of the random variable X , the number of the trial on which the k th success occurs, is

$$b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}, \quad x = k, k+1, k+2, \dots$$

Example 5.14: In an NBA (National Basketball Association) championship series, the team that wins four games out of seven is the winner. Suppose that teams A and B face each other in the championship games and that team A has probability 0.55 of winning a game over team B .

- (a) What is the probability that team A will win the series in 6 games?
- (b) What is the probability that team A will win the series?
- (c) If teams A and B were facing each other in a regional playoff series, which is decided by winning three out of five games, what is the probability that team A would win the series?

Solution: (a) $b^*(6; 4, 0.55) = \binom{5}{3} 0.55^4 (1 - 0.55)^{6-4} = 0.1853$

(b) $P(\text{team } A \text{ wins the championship series})$ is

$$\begin{aligned} & b^*(4; 4, 0.55) + b^*(5; 4, 0.55) + b^*(6; 4, 0.55) + b^*(7; 4, 0.55) \\ &= 0.0915 + 0.1647 + 0.1853 + 0.1668 = 0.6083 \end{aligned}$$

(c) $P(\text{team } A \text{ wins the playoff})$ is

$$\begin{aligned} b^*(3; 3, 0.55) + b^*(4; 3, 0.55) + b^*(5; 3, 0.55) \\ = 0.1664 + 0.2246 + 0.2021 = 0.5931. \end{aligned}$$

If repeated independent trials can result in a success with probability p and a failure with probability $q = 1 - p$, then the probability distribution of the random variable X , the number of the trial on which the first success occurs, is

$$g(x; p) = pq^{x-1}, \quad x = 1, 2, 3, \dots$$

Example 5.15: For a certain manufacturing process, it is known that, on the average, 1 in every 100 items is defective. What is the probability that the fifth item inspected is the first defective item found?

Solution: Using the geometric distribution with $x = 5$ and $p = 0.01$, we have

$$g(5; 0.01) = (0.01)(0.99)^4 = 0.0096.$$



Example 5.16: At a “busy time,” a telephone exchange is very near capacity, so callers have difficulty placing their calls. It may be of interest to know the number of attempts necessary in order to make a connection. Suppose that we let $p = 0.05$ be the probability of a connection during a busy time. We are interested in knowing the probability that 5 attempts are necessary for a successful call.

Solution: Using the geometric distribution with $x = 5$ and $p = 0.05$ yields

$$P(X = x) = g(5; 0.05) = (0.05)(0.95)^4 = 0.041. \quad \blacksquare$$

Quite often, in applications dealing with the geometric distribution, the mean and variance are important. For example, in Example 5.16, the *expected* number of calls necessary to make a connection is quite important. The following theorem states without proof the mean and variance of the geometric distribution.

$$P(X = x) = (1 - p)^{x-1}p$$

The mean and variance of a random variable following the geometric distribution are

$$\mu = \frac{1}{p} \text{ and } \sigma^2 = \frac{1-p}{p^2}.$$

